SymPy Bug Card

Bug 62: Series of $(x^3)^{1/3}$ Near -1 Uses the Wrong Principal Branch

Task. Expand the principal cube root near a negative real point.

$$\operatorname{series}\Big((x^3)^{1/3}, x, -1, 2\Big)$$

SymPy's answer.

$$\operatorname{series}\Big((x^3)^{1/3}, x, -1, 2\Big) = x$$

Correct answer.

$$\lim_{x\to -1}(x^3)^{1/3} = e^{i\pi/3} \neq -1$$

Explanation. For negative real x near -1, x^3 is negative real, so the principal cube root has argument $\pi/3$. Thus $(x^3)^{1/3} = (-x)e^{i\pi/3}$ on that side, while SymPy's series x gives the real value -1 at the center.

Likely root cause. In `sympy/core/power.py`, `Pow._eval_nseries()` applies `powdenest(force=True)`, which rewrites `((t - 1)**3)**(1/3)` to `t - 1` after shifting the expansion point. That forced denesting loses the principal-branch phase; this is a new instance of a known failure mode.

Artifact (GitHub). `bug-report/bug-062-series-of-x-3-1-3-near-1-uses-the-wrong-principal-branch/`